Department of Computer Science & Engineering
The University of Texas at Arlington

Project Charter
CSE 4316: Senior Design I
Summer 2025

IGVC
MavTruck

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Revision History

Revision	Date	Author(s)	Description
0.1	07.01.2025	KS,TT,BB,RR,AK,KL	Initial document draft
0.1	07.05.2025	RR	Added Assumption, Risk
0.1	07.07.2025	RR	Added SRS
0.2	07.08.2025	KS,RR	Added background and system overview sections, and Related Work
0.3	07.09.2025	TT, BB, RR	Added methodology and value proposition
0.4	07.10.2025	KL, KS	Completed constraints and development milestones
0.6	07.10.2025	AK,RR	Completed Problem Statement and added Closeout Materials
0.7	07.10.2025	AK	Completed System Overview
1.0	07.11.2025	KS,TT,BB,RR,AK,KL	Final version for review

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1 Problem Statement

The problem addressed in this project is the need for a fully autonomous ground vehicle capable of safely navigating an outdoor environment without human control. The vehicle must detect and avoid obstacles, interpret lane markings, and follow a designated route using onboard sensors and computing resources. These requirements reflect real-world challenges in autonomous mobility systems, where reliability, environmental awareness, and consistent decision-making are critical. Our goal is to design, build, and validate a robotic platform that can operate independently in dynamic outdoor conditions while meeting the performance expectations set by the Intelligent Ground Vehicle Competition (IGVC).

2 Methodology

To address the problem of autonomous outdoor navigation, we will design and implement a modular robotic system composed of several integrated subsystems. The vehicle will include a reinforced chassis, a motor control system for differential steering, a sensor suite for environmental perception, and a navigation module capable of path planning and obstacle avoidance. GPS, LiDAR, and vision sensors will be incorporated to support localization and situational awareness. A central software controller will fuse sensor data, generate movement commands, and execute decision-making algorithms. Each subsystem will be developed and tested individually, then incrementally integrated into the full platform. Throughout the build process, we will use iterative development methods that include subsystem prototyping, simulation-based testing, and continuous refinement based on field performance. This methodology ensures that the final vehicle is reliable, safe, and capable of meeting IGVC navigation requirements.

3 Value Proposition

The Intelligent Ground Vehicle Competition (IGVC) project represents a collaborative initiative led by our student team to design, develop, and implement a fully autonomous ground vehicle. This project offers a valuable opportunity for stakeholders including industry sponsors, faculty, and the university to actively engage in the advancement of autonomous systems and intelligent transportation technologies. By exploring key areas such as robotics, computer vision, and embedded systems, the project contributes to innovation in emerging fields while enhancing the institution's presence in cutting-edge engineering domains.

Participation in IGVC allows us, as students, to bridge the gap between academic knowledge and practical application. Working in a multidisciplinary team environment, we gain critical hands-on experience in hardware integration, software development, and systems engineering. Beyond technical skills, the project cultivates essential competencies such as teamwork, project management, and creative problem solving preparing us to enter the workforce as capable and well-rounded professionals.

This initiative also strengthens collaboration between academia and industry, opening avenues for mentorship, research, and long-term partnership. The success of the IGVC project not only elevates the university's reputation but also supports the broader mission of driving innovation in autonomous mobility. Ultimately, this project stands as a testament to future-focused learning, applied engineering, and meaningful collaboration that benefits all involved.

4 Development Milestones

This list of core project milestones should include all major documents, demonstration of major project features, and associated deadlines. Any date that has not yet been officially scheduled at the time of preparing this document may be listed by month.

Provide a list of milestones and completion dates in the following format:

- Project Charter first draft - June 2025

- System Requirements Specification - July 2025
- Architectural Design Specification - July 2025
- Demonstration of Robot Chassis and Motor Integration - August 2025
- Detailed Design Specification - September 2025
- Demonstration of Obstacle Detection and Avoidance - September 2025
- Demonstration of Sensor Integration - October 2025
- CoE Innovation Day poster presentation - November 2025
- Demonstration of Lane Detection - November 2025
- Demonstration of Path Planning and GPS Waypoint Navigation - November 2025
- Demonstration of Final Integrated System (Autonomous Mode with Obstacle Avoidance, Lane Following, and GPS Waypoint Navigation) - December 2025
- Final Project Demonstration (Self-Driving Car with full autonomy) - December 2025

5 Background

As undergraduate students entering the rapidly evolving world of autonomous systems, we recognize that the future of transportation lies in intelligent self-navigating vehicles. From delivery robots to self-driving cars, autonomous mobility is no longer a futuristic idea it is an ongoing technological revolution. To become competitive engineers in this new age, we believe it is essential to build a strong foundation in robotics, sensor integration, and real-world problem solving.

The Intelligent Ground Vehicle Competition (IGVC) provides an ideal platform for students like us to get a head start in this field. It challenges teams to design and build a fully autonomous ground vehicle that can navigate complex outdoor environments while detecting obstacles, following lanes, and reaching GPS-defined waypoints. More than just a competition, IGVC serves as a practical classroom where we apply what we have learned in embedded systems, algorithms, and controls to a real-world engineering challenge.

Unlike industry labs or commercial projects that often use high-end systems, our approach prioritizes simplicity and cost-effectiveness. By working with accessible tools such as GPS, IMUs, ultrasonic sensors, and microcontrollers, we learn to solve problems within real-world constraints. This not only helps us understand the core principles of autonomous navigation, but also prepares us for working in environments where resources are limited and creativity is essential.

Although our ultimate goal is to compete successfully in the IGVC, the journey itself equips us with hands-on experience, teamwork, and a mindset of innovation, all of which are crucial as we enter an industry that values both technical depth and adaptability.

6 Relat ed Work

The Intelligent Ground Vehicle Competition (IGVC) is an intercollegiate event where university teams design and demonstrate autonomous ground vehicles, many of which utilize inexpensive sensors and microcontroller-based systems for navigation and obstacle avoidance [1]. While previous teams at the University of Texas at Arlington successfully participated in past competitions, our project takes a different approach by emphasizing simplicity and affordability. Instead of relying on complex middleware like ROS or expensive sensors such as LiDAR, we are developing a lightweight system using GPS, an IMU, ultrasonic sensors, and a microcontroller to achieve reliable navigation and obstacle detection.

Our project draws inspiration from existing research on low-cost GPS and IMU fusion methods for outdoor navigation. In particular, the approach presented in [2] demonstrates how Kalman filters can combine GPS and IMU data to achieve decimeter-level accuracy in unstructured environments. By adapting similar techniques, we aim to implement an affordable and efficient solution for reliable localization in the IGVC course.

Another relevant area is obstacle detection using ultrasonic sensors. In [3], Ma (2023) proposed a system integrating multiple ultrasonic sensors with real-time processing to improve obstacle avoidance in dynamic environments. Together, these studies provide a foundation for our simplified design tailored to meet the specific requirements of the IGVC competition.

7 System Overview

Our plan is to build a four-wheeled vehicle with rear-wheel drive. The back wheels will be powered by 2 wheelchair motors and a standard lead-acid battery will be utilized to power these 2 motors. The chassis is made of t-slot aluminum bars connected with T/L brackets and screws. This creates the sturdy and durable base we need to hold the weight of the motors, battery, and payload. The body of the vehicle is going to be designed to resemble a Cybertruck. We are thinking the sides of the body can be 3D printed and the top will be metal. The top will be screwed to the top and will be designed to lift for easy access to internal components and wiring. A major goal is the ability to easily take it apart to fit in a checked

luggage

Since the goal of the vehicle is to autonomously guide itself through a series of obstacles, we have decided on 4 components to gather information about the environment. We will use a camera to detect white lines and the color of objects on the path, an ultrasonic sensor to detect objects in front of or around the vehicle, a GPS to detect positioning of the vehicle, and an IMU to detect the speed/acceleration of the vehicle. These components will work together to give a complex picture of the environment and allow navigation decisions to be made. To process this data, we are in contact with our lab manager to utilize a Jetson ORION that has been used in the past. This system will handle vision processing, object detection, and decision-making for speed and turning. The final part on our project is a mandatory stop button. This button needs to be sticking up from the base and needs to have a wireless activation as well. When this button is pressed, the vehicle will need to power down completely, in the case of an accident.

8 Roles & Responsibilities

The IGVC 2025 project includes a diverse set of stakeholders and team members, each contributing to the overall goal of designing, building, and programming a fully autonomous ground vehicle for competition.

8.1 Stakeholders:

- Primary Stakeholder: Dr. McMurrough and Dr. Conly will serve as the primary point of contact for all technical reviews, budget decisions, and milestone approvals.
- UTA CSE Department: Provides project funding, laboratory space, and access to departmental makerspace resources needed for development and testing.
- UTA College of Engineering: As the overarching academic sponsor, the College provides funding, lab facilities, and senior design evaluation.
- Future IGVC Teams: The next cohort of UTA senior design students will inherit and build upon this project. Therefore, long-term maintainability, modularity, and documentation are key deliverables for this stakeholder group.

8.2 Team Members and Responsibilities:

- Alexander Kowis: Embedded Systems Engineer, Communication Interface, Networking, Documentation Reporting
- Brian Biju: Power Systems, Motor Driver Coding, PCB Integration, Documentation Reporting
- Thanh Tran: Lane Detection, Object Tracking, RealSense Tuning, Documentation Reporting
- Khanh Lam: External Liaison, Mechanical Integration, Documentation Reporting
- Kopil Sharma: Power Systems, Motor Driver Coding, Obstacle Avoidance Algorithms, Documentation Reporting
- Rezwana Roza: GPS Integration, Path Planning PCB Integration, Documentation Reporting

All members will contribute collaboratively across development sprints and be cross-trained where possible to ensure redundancy and system-wide understanding.

Scrum Agile Team Structure: Our team will follow Agile Scrum practices with 2-week sprint cycles. The Scrum Master role will rotate among team members each sprint to give everyone experience in project coordination, retrospective facilitation, and task tracking. This rotation also ensures that no single member is burdened with all organizational duties for the duration of the project.

Product Owner: The Product Owner role will also be assigned within the team, rotating every major project milestone. The Product Owner is responsible for defining project goals, maintaining the product backlog, prioritizing tasks based on technical needs and competition requirements, and ensuring that sprint work aligns with the overall system objectives. The Product Owner communicates expectations, clarifies requirements, and helps guide decision-making throughout development.

This structure promotes both accountability and collaboration, ensuring a well-rounded team dynamic and consistent project momentum.

9 Cost Proposal

Since this project has been developed over multiple semesters by different teams, many of the major components and hardware have already been acquired. As a result, the current team is able to reuse several parts from previous builds, which significantly reduces overall costs.

9.1 Preliminary Budget

Item	Unit Cost	Quantity	Total Cost
Jetson AGX Orin 64GB Module	$700.00	1	$700.00
Lidar Unit	$560.00	1	$560.00
Sabertooth Motor Controller	$129.00	2	$258.00
24V DC Motors	$100.00	2	$200.00
24V Li-Fepo4 Battery Pack	$150.00	1	$150.00
Structural Frame Components	$5.99	20 pk	$119.80
USB Cameras	$45.00	2	$90.00
TM4C123GH6PM Microcontrollers (Sender + Receiver)	$24.00	3	$72.00
Miscellaneous Hardware (wires, connectors, mounts)	$10.00	6 pk	$60.00
HC-160A SC Motor Drivers (optional pair)	$25.00	2	$50.00
DC-DC Converters (5V, 19V)	$20.00	2	$40.00
LCD screen	$40.00	1	$40.00
Wire/Cable Packs	$10.00	4	$40.00
Wheel Encoders	$15.00	2	$30.00
STM32F446RE	$15.00	2	$30.00
T-Slot Bar Connectors (24 pcs)	$9.99	3 pk	$29.97
3D Printed Mounts	$5.00	5	$25.00
GPS Module	$20.00	1	$20.00
LED Tower Indicator	$18.00	1	$18.00
NRF24L01+ RF Modules	$6.00	2	$12.00
Emergency Stop Button	$12.00	1	$12.00
Total Project Cost			**$2,556.77**

Table 1: Preliminary budget estimate for IGVC project

9.2 Current & Pending Support

Our Senior Design Project for the Intelligent Ground Vehicle Competition (IGVC) is currently supported by an $800 budget provided by the Computer Science and Engineering (CSE) department. While this

covers some of our basic expenses, we're actively looking for additional funding to enhance our project. We plan to apply for university-sponsored undergraduate research grants, seek donations from alumni or community partners, and explore department-level discretionary funds. With extra financial support, we'll be able to upgrade key components, improve system reliability, and increase our chances of success at the competition.

10 Facilities & Equipment

Our primary workspace will be Room 203 in the Engineering Lab, located in the Engineering and Research Building at UTA. The team meets here during scheduled Friday lab sessions to collaborate and make consistent progress. The lab also provides flexible access hours, supporting ongoing development of both hardware and software components.

For tools and equipment, we will need soldering kits for assembling electronic circuits and a metal cutting machine to shape the robot's base frame, ensuring strong connections and structural integrity. To simulate the IGVC competition environment, we will require a large open area where we can place barrels, traffic cones, and marked lanes for navigation testing and data collection.

Setting up this testing area will likely require permission and space reservations, coordinated through UTA's Parking and Transportation Department. All essential components and materials will be purchased following consultation with Professor Conly and agreement among team members.

We will also use additional resources such as the Tiva microcontroller for embedded control and personal laptops for programming, data processing, and overall software development throughout the project.

11 Assumptions

The following list contains critical assumptions related to the implementation and testing of the project.

- A suitable outdoor testing area with a reliable GPS signal will be available
- All hardware components will be delivered on schedule
- The vehicle's power system will provide adequate and stable power to support all onboard sensors and processors
- The university network infrastructure and internet access will support software updates, remote repository access, and real-time collaboration.
- All sensors (ultrasonic sensors, cameras, GPS, IMU) will integrate seamlessly and provide accurate, noise-reduced data for perception and localization tasks.
- IGVC rules and specifications will remain stable during the development cycle.

12 Constraints

The following constraints represent external factors and limitations that influence the design, development, and testing of our IGVC autonomous ground vehicle:

- New environment: Our primary focus is to build the vehicle with Orion Nano which nobody in our group has worked on. This might be a challenge because of it being more python heavy rather than C which we are most familiar with.
- Summer-Fall team: Summer-Fall teams usually have strict deadlines and don't get any break in between semesters. The summer term is very short as well. So it might not give us enough time to incorporate all the features.

- Hardware Delivery and Lead Times: We are building up in the previous teams work but there are several aspects that need to be changed and the components that needs to be added might not be available right when we want it.
- Team Member Availability: Since team members have varied class schedules, internships, and work commitments, coordination and sprint velocity may be impacted by limited overlapping work hours.
- Computer engineer team: Everyone in our team is a computer engineering major. While it might be a benefit since all of us will be in sync but it might also be a problem since all of us are good with hardware rather than software.

These constraints guide our planning process and emphasize the importance of early procurement, efficient testing cycles, and robust task management.

13 Risks

The following high-level risk census contains identified project risks with the highest exposure. Mitigation strategies will be discussed in future planning sessions.

Risk description	Probability	Loss (days)	Exposure (days)
Functionality of navigation logic	0.50	20	10
Subsystem integration issues	0.20	5	4
Battery overheating	0.50	5	1.5
Loose or damaged connectors	0.30	4	2.0

Table 2: Overview of highest exposure project risks

14 Documentation & Reporting

14.1 Major Documentation Deliverables

14.1.1 Project Charter

The Project Charter outlines the initial scope, goals, responsibilities, and risks for the IGVC 2025 project. This document is collaboratively maintained and updated by the entire team. The initial draft was completed on July 11, 2025, following team formation and early planning. Revisions to the charter will occur at major project milestones, such as the completion of system architecture or a significant shift in project scope or resource availability. The final version will be delivered by August 2025.

14.1.2 System Requirements Specification

The System Requirements Specification (SRS) for the IGVC project defines the functional and non-functional requirements of the autonomous ground vehicle to ensure successful participation in the Intelligent Ground Vehicle Competition (IGVC). The document will be created, maintained, and updated collaboratively using Overleaf in LaTeX format.

14.1.3 Architectural Design Specification

The Architectural Design Specification (ADS) outlines the system-level architecture of the autonomous vehicle, including hardware subsystems, software modules, communication interfaces, and data flow. It will be maintained through a collaborative Git-based workflow with version tracking.

14.1.4 Detailed Design Specification

The Detailed Design Specification (DDS) provides in-depth information on the implementation of hardware and software systems, including sensor configurations, control algorithms, motor interfaces, and PCB schematics. This document evolves alongside subsystem development.

14.2 Recurring Sprint Items

14.2.1 Product Backlog

Items are added to the product backlog based on the future requirements of the project. These tasks are identified through team discussions and collaborative decision-making, with updates managed by Kopil. The product backlog will be maintained and regularly updated in Github, with team notifications sent via Discord to ensure everyone stays informed.

14.2.2 Sprint Planning

Sprint plans will be developed during our weekly team meetings on Discord. We estimate that approximately 8 sprints will be required to complete the project from start to finish.

14.2.3 Sprint Goal

Sprint goals will be determined by the team based on project progress, with input from the customer during planning. For important decisions, the team consults project supervisors through meetings or discussions.

14.2.4 Sprint Backlog

During sprint planning, the team collaboratively decides which items from the product backlog should be included in the upcoming sprint. Decisions are based on task priority, team availability, and current project goals.

14.2.5 Task Breakdown

Tasks from the sprint backlog are voluntarily claimed by team members during sprint planning meetings. Each member selects tasks based on their strengths and interests. For larger or more complex tasks, multiple team members may share responsibility to allow for collaboration and manageable workload. Task ownership and assignments are documented in GitHub to ensure transparency.

14.2.6 Sprint Burn Down Charts

There is no requirement for generating a burn down chart in this class or project.

14.2.7 Sprint Retrospective

Retrospectives of sprints will take place at the end of each sprint during our regular team meeting. The team will reflect on three key point, what went well, what did not go well and what can be improved.

Each team member will contribute feedback, and notes will be recorded in a shared Google Doc.

14.2.8 Individual Status Reports

Each member will submit a brief sprint report in canvas at the end of the sprint usually happening every 3 weeks. The report will include:

- Tasks completed
- Time spent on each task
- Challenges faced

- Planned tasks for the next sprint
- Ratings for the team members

14.2.9 Engineering Notebooks

No engineering notebook is being used. Online whiteboards and google docs are used to plan and divide tasks and update information.

14.3 Closeout Materials

14.3.1 System Prototype

The team will develop a prototype autonomous vehicle using a wheelchair motor base, GPS, IMU, ultrasonic sensors, and a microcontroller to test core navigation and obstacle detection functionalities. The final system will integrate all hardware and software required to comply with IGVC rules and successfully compete. There will be no Prototype Acceptance Test (PAT) or Field Acceptance Test (FAT).

14.3.2 Project Poster

Our project poster will be designed on a tri-fold board measuring 36 inches by 48 inches. It will provide a clear and visually engaging summary of the design and development of the autonomous vehicle, including the integration of GPS, IMU, ultrasonic sensors, and microcontroller-based control.

14.3.3 Web Page

A project web page will be created to showcase the design process, system architecture, and testing results of the autonomous vehicle. It will include updates on project milestones, images and videos of the prototype, and a section for future teams to access relevant documentation and resources.

14.3.4 Demo Video

Our team will create short 2-3 minutes demo videos highlighting the vehicle's design, key features, and testing outcomes. These videos will also display project challenges and solutions.

14.3.5 Source Code

The project's source code will be maintained in a Git repository to ensure proper version control and team collaboration. The repository will have a main branch for stable code and separate feature branches for development tasks. Team members will work on individual branches and merge changes into the main branch after review and testing. A designated repository manager will oversee merge requests and resolve any conflicts to maintain a clean and functional codebase.

14.3.6 Source Code Documentation

Our team will use Doxygen to generate clear and organized documentation for all source code.

14.3.7 Hardware Schematics

The team will design custom printed circuit boards (PCBs) to integrate key components such as the GPS module, IMU, ultrasonic sensors, and motor controller. Schematics will include detailed wiring diagrams for power distribution, sensor connections, and microcontroller interfaces.

14.3.8 CAD files

At this time, the team does not plan to create any custom CAD files.

14.3.9 Installation Scripts

The project will include basic installation instructions to help future teams set up and configure the autonomous vehicle system. Our resources will be included in the Git repository to ensure the system can be replicated and extended by future cohorts.

14.3.10 User Manual

A digital user manual will be created to provide future teams with clear instructions for operating, maintaining, and modifying the autonomous vehicle. The manual will include detailed sections on hardware setup, system calibration (for GPS and sensors), and troubleshooting common issues. Our team will also provide supplementary videos to demonstrate key features and processes for ease of understanding

References

[1] Intelligent Ground Vehicle Competition, "Official igvc website," http://www.igvc.org/, accessed: 2025-07-03.

[2] T. Smith and A. Jones, "Autonomous navigation in outdoor environments using gps and imu fusion," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 1234–1240.

[3] J. Ma, "A real-time obstacle detection framework using ultrasonic sensor arrays in autonomous robots," *Highlights in Science, Engineering and Technology*, vol. 71, pp. 68–78, 2023.